AL-2001
Artificial Intelligence
Lab 12

Objectives:
- Matplotlib

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. Do not copy from any source otherwise you will be penalized with negative marks.
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

Task 1:

Task: Create a simple line plot.

Generate a NumPy array 'x' with values from 0 to 10.

Create another array 'y' with values representing the square of each element in 'x'.

Plot 'x' against 'y' using a blue line.

Add labels to the x-axis and y-axis, and a title to the plot.

Display the plot.

Task 2:

Task: Create a scatter plot with color and size.

Generate two arrays 'x' and 'y' with random values.

Create a third array 'sizes' with random sizes for each point.

Plot a scatter plot of 'x' and 'y' where each point has a size proportional to the values in 'sizes'.

Use a colormap for colors.

Add labels to the x-axis and y-axis, and a title to the plot.

Display the plot.

Task 3:

Task: Create a bar chart with error bars.

Generate an array 'categories' with some labels.

Create two arrays 'values' and 'errors' with corresponding values and error values.

Plot a bar chart with error bars for each category.

Customize the appearance of the bars and error bars.

Add labels to the x-axis and y-axis, and a title to the plot.

Display the plot.

Task 4:

Task: Create a histogram with multiple bins and overlays.

Generate a random dataset with 1000 values.

Create a histogram with 20 bins, using a specific color and edge color.

Overlay a second histogram with 30 bins using a different color.

Add labels to the x-axis and y-axis, and a title to the plot.

Display the plot.

Task 5:

Task: Create a complex plot with subplots.

Generate three datasets (arrays) with different mathematical functions.

Create a figure with three subplots arranged in a 2x2 grid.

Plot each dataset in a different subplot with appropriate labels and titles.

Customize the appearance of each subplot.

Add an overall title to the entire figure.

Display the plot.

You need to done with your exercise within given time.